# An Econometric Analysis of Rent Control

Edgar O. Olsen

*University of Virginia*

Rent control affects the allocation of resources and the distribution of well-being. In New York City in 1968, it is estimated that occupants of controlled housing consumed 4.4 percent less housing service and 9.9 percent more nonhousing goods than they would have consumed in the absence of rent control. The resulting increase in their real income was 3.4 percent. Poorer families received larger benefits than richer families. The cost of rent control to landlords was twice its benefit to their tenants. The estimates are produced within the framework of a simple general equilibrium model; the data are on thousands of families and their apartments.

> It is perfectly astounding to find that the quantity and quality of analytical materials on rent control are wholly incommensurate with the social and economic importance of the problem. [Grebler 1952, p. 464]

Rent control has affected patterns of consumption and well-being for at least fifty years; yet empirical evidence on the economic effects of this form of government regulation is scanty and crude.[1] The primary pur-

This paper has been presented at the 1970 winter meetings of the Econometric Society, the urban economics workshop at the University of Chicago, and seminars at Cornell University, Northwestern University, State University of New York at Stony Brook, University of Texas, University of Virginia, and Virginia Polytechnic Institute and State University. I wish to thank the participants in these seminars for many helpful comments. I am also grateful to Edgar K. Browning, Peter Mieszkowski, Larry Orr, and the referees for more detailed criticisms and suggestions. Finally, I would like to recognize my great debt to Joseph S. DeSalvo for numerous productive conversations during the conception of this paper.

[1] To many American economists, rent control is but a dim memory. It had ended in most parts of the United States by 1951. However, in the rest of the world it is still very much alive. For example, rent controls covered about one-half of privately rented dwellings in Britain in 1964 (Needleman 1965, p. 165), one-third of all dwelling units in Sweden in 1960 and probably a similar proportion in 1965 (Stahl 1967, p. 74),

poses of this paper are to provide a general method for answering certain questions about rent control and to answer these questions for New York City in 1968.

The following questions will be answered. What difference has rent control made in the consumption of housing service and nonhousing goods by occupants of controlled housing? What is the mean benefit of rent control to tenants occupying rent-controlled apartments? What is the total benefit to all families living in controlled units? How does the total benefit to these families compare with their total income? Who bears the cost of rent control, and what is the magnitude of this cost? Are the occupants of controlled housing poorer than the occupants of uncontrolled housing? Among the set of families living in controlled housing, how does the size of tenant benefit vary with household income, family size, and the age, sex, and race of the head of the household?

## I. Theoretical Framework

The questions posed will be answered with the help of a simple economic model. Assume that there are two goods—housing service and nonhousing goods—and three markets—the controlled market for housing service, the uncontrolled market for housing service, and the market for nonhousing goods. Assume that the latter two markets are perfectly competitive and that we observe both markets in long-run equilibrium. Assume that the long-run supply curves in both markets are perfectly elastic.[2] Assume that the long-run supply price of housing service in the uncontrolled market is the same as it would have been in the absence of rent control.

Before proceeding to use these assumptions to derive formulas for answering the questions posed, it is desirable to define the phrase "housing service" with some care. Housing service is an unobservable good emitted in some quantity by each dwelling unit during each period of time. It is the one and only thing in a dwelling unit to which consumers attach value. Intuitively, the quantity of housing service emitted by a dwelling unit can be thought of as an index of all its attributes, including both space and quality. Since it is assumed that the uncontrolled housing market is perfectly competitive, we know that in equilibrium each unit of housing

and 18 percent of the population of Mexico City in 1964 (Aaron 1966, pp. 316–17). Even in the United States, there were 1.3 million rent-controlled apartments in New York City in 1968 and only 0.7 million federal public-housing units in the entire country (U.S. Bureau of the Census 1970, p. 682). Furthermore, rent control is making a comeback in the United States. In 1969 Boston and Hartford implemented mild forms of rent control while New York City placed an additional 375,000 apartments under a rent-control law less severe than that applied to previously controlled apartments (Fenton 1969, p. 21; Shipler 1970, p. 35).

[2] This assumption is consistent with the finding of Muth (1960; pp. 42–46), but de Leeuw and Ekanem (1971, p. 814) find price elasticities of long-run supply between 0.3 and 0.7.

service sold on this market sells for the same price. Hence, if we observe one apartment renting for \$200 per month and another for \$100 per month on the uncontrolled housing market, then we say that the first apartment emits twice the quantity of housing service per time period as the second.[3]

Let us now use the assumptions made to derive formulas which, when combined with the available data and predictions based on these data, will yield the answers that we seek. In figure 1, $D$ is the demand curve for housing service of some tenant of a rent-controlled dwelling unit, holding the price of nonhousing goods and money income constant. If the uncontrolled market price of housing service were $P_m{}^h$ and if the tenant were to buy housing service on the uncontrolled market, then he would occupy a dwelling emitting $Q_m{}^h$ units of housing service per time period and pay a "rent" of $P_m{}^h\ Q_m{}^h$. Finally, let us suppose that under rent control the tenant occupies a dwelling unit emitting $Q_c{}^h$ units of housing service per time period and pays a rent of $P_c{}^h\ Q_c{}^h$ for this dwelling unit.

The relative magnitudes of the variables depicted in figure 1 do not represent the situation of every occupant of a controlled apartment. For example, a significant minority of the occupants of controlled apartments live in better housing than they would have occupied in the absence of rent control (that is, $Q_c{}^h > Q_m{}^h$). Nevertheless, the empirical evidence will show that the relationships between variables depicted in figure 1 are typical of families living in controlled housing.

It is perhaps not obvious why the occupant of a controlled apartment would consume a smaller quantity of housing service at the lower controlled price. The explanation is as follows. The price-quantity combination $(P_c{}^h, Q_c{}^h)$ is preferred to the tenant's best alternative on the uncontrolled market $(P_m{}^h, Q_m{}^h)$ because it permits a sufficiently greater consumption of nonhousing goods to offset the smaller consumption of housing service. Furthermore, the owner of this controlled apartment has no incentive to increase its flow of services. To do so would increase cost but not revenue. Finally, there is nothing in the operation of the controlled housing market which guarantees that this tenant can find an apartment renting for $P_c{}^h\ Q_d{}^h$ and yielding $Q_d{}^h$ units of housing service per time period. It is extremely difficult to obtain a controlled apartment, and the tenant might well judge the cost of finding a better bargain on the controlled market to be greater than the gain. If he does search for a better bargain, he has no incentive to limit his search to apartments with price-quantity combinations on his demand curve. In short, there is no theoretical reason to believe that all occupants of controlled apartments

[3] Muth (1960) has developed and tested a competitive theory of the housing market using this definition of housing service.

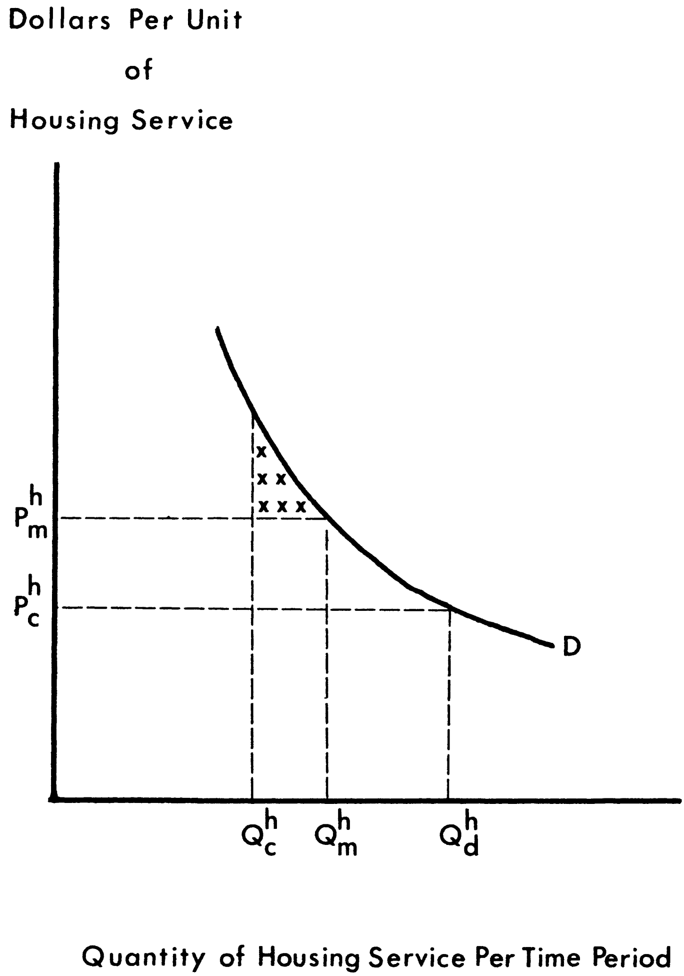


FIG. 1.—The effects of rent control on the occupant of a controlled apartment and his landlord.

are on their demand curves, and the evidence to be presented is inconsistent with this hypothesis.

In the case depicted in figure 1, the tenant consumes

$$100\ [(P_m{}^h\, Q_m{}^h - P_m{}^h\, Q_c{}^h)/P_m{}^h\, Q_m{}^h] \text{ percent} \tag{1}$$

less housing service than he would have consumed in the absence of rent control. On the other hand, he spends

$$P_m{}^h\, Q_m{}^h - P_c{}^h\, Q_c{}^h$$

more on nonhousing goods. Hence, as a result of rent control, he consumes a greater quantity of nonhousing goods. To be precise, he consumes

$$100\ [(P_m{}^n\, Q_c{}^n - P_m{}^n\, Q_m{}^n)/P_m{}^n\, Q_m{}^n] \text{ percent} \tag{2}$$

more nonhousing goods, where $P_m{}^n$ is the market price per unit of nonhousing goods, $Q_m{}^n$ is the quantity of nonhousing goods that this consumer would purchase in the absence of rent control, and $Q_c{}^n$ is the quantity of nonhousing goods that he consumes under rent control. Since there are only two goods, everything that the consumer does not spend on housing service is spent on nonhousing goods. Hence,

$$P_m{}^n\, Q_c{}^n = Y - P_c{}^h\, Q_c{}^h \tag{3}$$

and

$$P_m{}^n\, Q_m{}^n = Y - P_m{}^h\, Q_m{}^h, \tag{4}$$

where $Y$ is the tenant's income. Substituting (3) and (4) into (2), we find that the percentage change in the consumption of nonhousing goods by this tenant may be written as

$$100\ [(P_m{}^h\, Q_m{}^h - P_c{}^h\, Q_c{}^h)/(Y - P_m{}^h\, Q_m{}^h)]. \tag{5}$$

We can calculate the proportional change in the total quantities of housing service and nonhousing goods for a set of families by formulas (6) and (7):

$$\frac{\sum_{i=1}^{k} Q_{c,i}{}^h - \sum_{i=1}^{k} Q_{m,i}{}^h}{\sum_{i=1}^{k} Q_{m,i}{}^h} = \frac{\sum_{i=1}^{k} P_m{}^h\, Q_{c,i}{}^h - \sum_{i=1}^{k} P_m{}^h\, Q_{m,i}{}^h}{\sum_{i=1}^{k} P_m{}^h\, Q_{m,i}{}^h}, \tag{6}$$

$$\frac{\sum_{i=1}^{k} Q_{c,i}{}^n - \sum_{i=1}^{k} Q_{m,i}{}^n}{\sum_{i=1}^{k} Q_{m,i}{}^n} = \frac{\sum_{i=1}^{k} P_m{}^h\, Q_{m,i}{}^h - \sum_{i=1}^{k} P_{c,i}{}^h\, Q_{c,i}{}^h}{\sum_{i=1}^{k} (Y_i - P_m{}^h\, Q_{m,i}{}^h)}, \tag{7}$$

where the subscript $i$ indicates the $i$th family. Notice that the derivation of these formulas requires the use of an implication of our assumptions, namely, that in a competitive market in equilibrium, all consumers pay the same price for the good.

Since the tenant has the option of living in uncontrolled housing, this change in his consumption pattern must make him better off. There is some amount of money $B$ that would make the tenant indifferent to the following two alternatives: remaining in his rent-controlled apartment during some time period, or accepting an unrestricted cash grant of $B$ dollars on the stipulation that he live in an uncontrolled apartment during the time period but could return to his controlled apartment at the end of the time period.[4] If we were to offer the tenant a cash grant of more than $B$ per time period, he would accept. If we were to offer him a cash grant of less than $B$, he would reject the offer and remain in his controlled apartment. I define $B$ to be the net benefit of rent control to the tenant.

Net tenant benefit is approximately equal to the excess of consumer's surplus at $(Q_c^h, P_c^h)$ over the consumer's surplus at $(Q_m^h, P_m^h)$.[5] The consumer's surplus at a point $(Q, P)$ is

$$\left(\int_0^Q D\right) - PQ.$$

Therefore, the excess of consumer's surplus at $(Q_c^h, P_c^h)$ over consumer's surplus at $(Q_m^h, P_m^h)$ is

$$\begin{aligned} &\left[\left(\int_0^{Q_c^h} D\right) - P_c^h Q_c^h\right] - \left[\left(\int_0^{Q_m^h} D\right) - P_m^h Q_m^h\right] \\ &= P_m^h Q_m^h - P_c^h Q_c^h + \int_{Q_m^h}^{Q_c^h} D \doteq B. \end{aligned} \tag{8}$$

Intuitively, the value of rent control to the tenant whose situation is depicted in figure 1 can be thought of as the sum of two components. First, the tenant benefits because he spends less money on housing service and hence is able to consume more nonhousing goods. The value of this component of net benefit is approximately

$$P_m^h Q_m^h - P_c^h Q_c^h, \tag{9}$$

the excess of housing expenditure in the absence of rent control over housing expenditure under rent control. Second, the tenant loses because he consumes less housing service. The value that the tenant places on a small change in the quantity of housing service consumed is approximately equal to the height of the demand curve times the magnitude of the change in quantity. Consequently, the loss that this tenant incurs by

[4] All moving costs are compensated for separately.

[5] The ideal way to estimate net tenant benefit is by reference to an indifference map estimated with data on individual responses to differences in prices and income. This approach is not used in this paper for several reasons. First, there is no published estimate of an indifference map in which the two arguments are housing service and nonhousing goods. Second, it is not possible to infer the parameters of such an indifference map from published estimates of the parameters of demand and expenditure functions in a way which permits the use of available data to estimate benefits. DeSalvo (1971) has made such inferences, but his formula for measuring benefits requires a measure of permanent income and my data contain no such information. Third, an indifference map cannot be estimated using my data. Fourth, to produce estimates of the parameters of the desired indifference map using another body of data is to produce another paper. Although we do not have an estimated indifference map, we do have other information about consumer behavior, such as the demand curve $D$ holding money income and the price of nonhousing goods constant. This curve permits the use of Marshallian consumer's surplus analysis to produce a formula for measuring net tenant benefit which is an approximation to the formula that would be obtained from an estimated indifference map. The sense in which it is an approximation has been investigated by a number of economists, including Hicks (1943).

consuming $Q_c^h$ rather than $Q_m^h$ units of housing service per time period is

$$\int_{Q_m^h}^{Q_c^h} D, \tag{10}$$

the negative of the area under the demand curve between $Q_m^h$ and $Q_c^h$. The sum of (9) and (10) is our approximation of net tenant benefit.

The expression (8) would be of little use in calculating net tenant benefit in the absence of a knowledge of the tenant's demand curve for housing service. Empirical evidence by Muth (1960), Reid (1962, p. 381), Lee (1964), and de Leeuw (1971) on the demand curve holding the price of nonhousing goods and money income constant suggests that the price elasticity of demand is between 0.7 and 1.7. Therefore, it is at least consistent with these empirical findings to assume that each tenant's demand curve for housing service has a constant price elasticity equal to 1.[6] Of all the assumptions about the demand curve for housing service consistent with the empirical evidence, this one is chosen because it leads to a particularly simple formula for calculating net tenant benefit. Specifically, since we assume that each tenant's demand curve is of the form

$$P^h = P_m^h Q_m^h / Q^h,$$

we will calculate net tenant benefit by the formula

$$P_m^h Q_m^h - P_c^h Q_c^h + P_m^h Q_m^h [\ln P_m^h Q_c^h - \ln P_m^h Q_m^h].^7 \tag{11}$$

The mean benefit to a set of families occupying controlled dwellings is obtained by computing the benefit to each family and calculating the mean of these numbers.[8]

In order to provide the tenant with this benefit, someone must bear the cost. The cost is borne primarily by present and past owners of the controlled apartment and by taxpayers who pay for the enforcement of the rent-control law. In the absence of rent control, the owner would receive

[6] Of course, I realize that these findings refer to aggregate demand curves and that an aggregate demand curve with constant price elasticity equal to 1 does not imply that each family's demand curve has unitary price elasticity. Indeed, I expect otherwise. For this reason, the estimate of the benefit to most families will err. Some estimates will be too high; others too low. Since even my discussion of distribution is only concerned with averages, this source of error is likely to be unimportant.

[7] $$\int_{Q_m^h}^{Q_c^h} D = \int_{Q_m^h}^{Q_c^h} P_m^h Q_m^h / Q^h = P_m^h Q_m^h [\ln Q_c^h - \ln Q_m^h]$$
$$= P_m^h Q_m^h [\ln P_m^h Q_c^h - \ln P_m^h Q_m^h].$$

[8] This is a consumer's surplus estimator of mean benefit. It differs from a consumers' surplus estimator obtained by substituting the sample means of $P_m^h Q_m^h$, $P_c^h Q_c^h$, and $P_m^h Q_c^h$ into formula (11). In principle, we want a consumer's surplus estimate, but in practice we must often settle for a consumers' surplus estimate because only aggregate data are available. Fortunately, data on individuals were available for this study.

$P_m^h Q_c^h$ for his apartment. Under rent control, he receives $P_c^h Q_c^h$. Hence, he would be willing to pay

$$P_m^h Q_c^h - P_c^h Q_c^h \tag{12}$$

per time period to have this apartment decontrolled, provided that he is unable to vary the quantity of housing service emitted by the apartment during this period of time.[9] This is our measure of the cost of rent control during a particular time period to the past and present owners of the controlled apartment. Like the measure of net tenant benefit, it is an approximation.

In principle, we would like to measure the cost incurred by present and past owners during a period of time as follows. We want to compare the situation under rent control with the situation that would have prevailed had rent control never gone into effect. Had rent control never been implemented, this apartment would have yielded its owner some revenue during each period of time and he would have incurred some cost of operating, maintaining, repairing, and altering the apartment. We shall call all of these "operating costs." Under rent control, the owner is likely to obtain some different revenue and to incur some different operating cost during each period of time. Let $Q_a^h$ be the quantity of housing service that would have been emitted by this particular apartment during some time period had rent control never gone into effect and $O_a$ be the operating cost that would have been incurred. Let $O_c$ be the operating cost that is actually incurred during the time period. The conceptually correct measure of the cost of rent control during this period of time to the present and past owners of the apartment is

$$(P_m^h Q_a^h - O_a) - (P_c^h Q_c^h - O_c),$$

which can be rewritten as

$$(P_m^h Q_c^h - P_c^h Q_c^h) + P_m^h (Q_a^h - Q_c^h) - (O_a - O_c).$$

Some operating costs are incurred on activities which affect only the flow of services that will be emitted by the dwelling during the same time period. Other costs are incurred on activities which affect the flow of services in later time periods. It is usually argued that $O_a$ is greater than

[9] Even if the controlled rent of an apartment is below its market rent, it does not follow that the present owner has been made worse off by the implementation of the rent-control law. For, suppose that the present owner bought the building in which the apartment is located while rent control was in effect and that at the time of the sale both the present owner and the previous owner correctly foresaw the future course of rent control. In this case, the purchase price of the building must have been just low enough so that the present owner could make a competitive rate of return on his investment. Otherwise, the present owner would have invested his money elsewhere. Thus, the cost calculated by formula (12) which shows up this period was actually incurred by previous owners. Of course, this argument is consistent with the present owner preferring decontrol to a continuation of controls.

$O_c$ during each period of time. If this were the case, then $Q_a{}^h$ would be greater than $Q_c{}^h$. Indeed, the difference between $Q_a{}^h$ and $Q_c{}^h$ is attributable solely to present and past differences between $O_a$ and $O_c$. Although $P_m{}^h$ $(Q_a{}^h - Q_c{}^h)$ and $(O_a - O_c)$ tend to cancel each other, they may differ in a particular time period, and hence formula (12) is an approximation to the conceptually correct measure. The conceptually correct measure is not used because the relevant data are not available.

If the only effect of rent control of interest to taxpayers is its administrative cost and if the only people concerned with the redistribution of consumption between a landlord and his tenant are these two people, then rent control must result in waste greater than zero. This waste is the excess of the sum of the loss to the landlord (12) and the cost of administering the program attributable to this particular dwelling unit over the gain to his tenant (11). Therefore,

$$W = P_m{}^h Q_c{}^h - P_m{}^h Q_m{}^h - P_m{}^h Q_m{}^h (\ln P_m{}^h Q_c{}^h - \ln P_m{}^h Q_m{}^h) + A. \tag{13}$$

In figure 1 the magnitude of the waste, net of administrative cost resulting from regulating the rent of this apartment, is the shaded area. The mean waste for a set of controlled apartments is obtained by computing the waste for each apartment and calculating the mean of these numbers.[10]

Looking back at the formulas derived in this section, we see that if we know each tenant's income, expenditure on housing under rent control, expenditure on housing in the absence of rent control, and the rent of his apartment on the uncontrolled market, then we can answer all of the questions posed in the introduction to this paper.

## II. Prediction of the Housing Expenditures of Families and the Market Rents of Dwellings

The empirical results of this paper are based on data from the 1968 New York City Housing and Vacancy Survey.[11] This survey collected many pieces of information for about 35,000 housing units and their occupants, including whether or not a housing unit is subject to rent control. It should be emphasized that the empirical results are based on data for individual families and housing units rather than on aggregates compiled from individual observations. Since the survey tells us directly the rent paid by the occupants of controlled housing and their income, we need only predict the housing expenditures of the occupants of controlled units in the absence of rent control and the market rents of their dwelling units in order to

[10] Again, this is not the same as substituting the sample means of $P_m{}^h Q_m{}^h$, $P_c{}^h Q_c{}^h$, $P_m{}^h Q_c{}^h$, and $A$ into formula (13) (see n. 8 above).

[11] A description of the sample design and an enumeration schedule containing the questions asked may be obtained from the U.S. Bureau of the Census.

answer the questions posed earlier. In essence, the method of prediction assumes that a family in a controlled unit would have spent in the absence of rent control the same amount on housing as the average expenditure on housing of families with the same characteristics who actually live in private, uncontrolled rental housing. Similarly, we assume that a controlled unit will rent on the uncontrolled market for the average of uncontrolled units with the same characteristics.

Specifically, assume that the stochastic model explaining differences in housing expenditure by different families occupying uncontrolled rental housing in New York City in 1968 was

$$\ln P_m^h Q_m^h = a_0 + a_1 \ln Y + a_2 \ln A + a_3 \ln N + a_4 R + a_5 S + u, \tag{14}$$

where $Y =$ annual income of the household, $A =$ age of head of household, $N =$ number of persons in the household, $R = 1$ if the head of household is nonwhite and 0 if white, $S = 1$ if the head of household is female and 0 if male, and $u \approx N(0, \sigma^2)$. We also assume that, given a sample of $T$ joint observations $(P_m^h Q_m^h)_t$, $Y_t$, $A_t$, $N_t$, $R_t$, and $S_t$ (where $t = 1, \ldots, T$) produced by the model (14), the successive disturbances $u_t (t = 1, \ldots, T)$ are mutually independent and that $Y_t$, $A_t$, $N_t$, $R_t$, and $S_t$ are nonstochastic. Our predictor of how much the occupants of a controlled unit would have spent on housing in the absence of rent control is

$$_e\hat{a}_0 + \hat{a}_1 \ln Y + \hat{a}_2 \ln A + \hat{a}_3 \ln N + \hat{a}_4 R + \hat{a}_5 S, \tag{15}$$

where the $\hat{a}$'s are least-squares estimators of the $a$'s in (14).[12] Using 4,063 families in uncontrolled housing for whom all of the necessary data were reported, the following results were obtained:[13]

$$\begin{aligned} \ln P_m^h Q_m^h = 4.071 &+ \underset{(41.2)}{0.351} \ln Y + \underset{(3.7)}{0.055} \ln A - \underset{(-0.6)}{0.007} \ln N \\ &- \underset{(-7.2)}{0.123} R + \underset{(7.9)}{0.114} S \qquad R^2 = .32. \quad s = .33 \end{aligned} \tag{16}$$

The $t$-scores are in parentheses beneath the estimated coefficients. For each of the 5,915 families in the sample occupying controlled housing, the value of the predictor (15) has been calculated.

12 This is not an unbiased predictor of either the conditional mean or the conditional median of the random variable $P_m^h Q_m^h$. However, because of the large sample size, the mean of the predictor is probably quite close to the conditional median of housing expenditure. The conditional mean of $P_m^h Q_m^h$ is probably about 6 percent greater than the conditional mean of my predictor. On these elusive matters, see Goldberger (1968).

13 It has been suggested that my estimate of the income elasticity is extraordinarily low. My estimate is low compared with estimates from studies in which the income variable is permanent income. However, my income variable is current income, and we expect the current income elasticity to be less than the permanent income elasticity.

Assume that the stochastic model explaining differences in the rents of different uncontrolled rental dwelling units in New York City in 1968 was

$$P_m{}^h\, Q_m{}^h = b_0 + \sum_{i=1}^{31} b_i X_i + u, \quad u \approx N\,(0, \sigma^2), \tag{17}$$

where the $X_i$ are as defined in table 1. We also assume that, given a sample of $T$ joint observations of the dependent variable and regressors produced by the model (17), the successive disturbances $u_t (t = 1, \ldots, T)$ are mutually independent and the regressors are nonstochastic. Our predictor of the market rent of an unfurnished controlled apartment is

$$\hat{b}_0 + \sum_{i=1}^{30} \hat{b}_i X_i \tag{18}$$

and of a furnished controlled apartment is

$$(\hat{b}_0 + \sum_{i=1}^{30} \hat{b}_i X_i)/(1 - \hat{b}_{31}), \tag{19}$$

where the $\hat{b}$'s are least-squares estimators of the $b$'s in (17).[14] Using 6,449 rental housing units on the uncontrolled market for which all the necessary data were reported, the results in table 1 were obtained.[15] For each of the 5,915 controlled dwelling units in our sample, the value of the predictor of market rent has been calculated.

## III. Empirical Results

We know the income of and rent paid by a sample of 5,915 families living in private, rental controlled housing, and we have predictions of how much each of these families would spend on housing and how much the dwelling unit that each occupies would rent for in the absence of rent control. The sample means of $P_c{}^h\, Q_c{}^h$, $P_m{}^h\, Q_c{}^h$, $P_m{}^h\, Q_m{}^h$, and $Y$ are \$999, \$1,405, \$1,470, and \$6,229, respectively.

Using formulas (6) and (7), we estimate that as a result of rent control the occupants of controlled units consumed 4.4 percent less housing service in total and 9.9 percent more nonhousing goods.[16] Using formula (11) to

[14] Larry Orr and James Stucker have pointed out quite correctly that the variable $X_{31}$ is correlated with the error term in (17). As a result, the $\hat{b}$'s are biased estimators of the $b$'s and hence the expected value of the predictor is not equal to the expected value of the random variable. I believe that the bias is small, but superior predictions could be obtained by estimating separate relationships for furnished and unfurnished apartments.

[15] For a more detailed analysis of housing expenditure and market rent functions based on data from the 1968 New York City Housing and Vacancy Survey, see DeSalvo (1970*b*).

[16] Had an estimate of the conditional mean rather than the conditional median of housing expenditure been used to predict how much each family would have spent on housing in the absence of rent control, the conclusions would have been that occu-

TABLE 1

Estimated Relationship between Market Rent and Housing Characteristics

| Regressors | Description of Regressors | Coefficients | t-Scores |
|---|---|---|---|
| $X_1$ | Number of bedrooms | 426.48 | 39.05 |
| $X_2$ | Number of other rooms | 294.84 | 19.98 |
| $X_3$ | 1 if dwelling sound and 0 if not sound | 534.96 | 6.29 |
| $X_4$ | 1 if dwelling deteriorating and 0 if not deteriorating | 370.68 | 4.02 |
| | ($X_3 = X_4 = 0$ if dwelling dilapidated) | | |
| $X_5$ | 1 if dwelling built in 1967–68 and 0 otherwise | 783.60 | 14.35 |
| $X_6$ | 1 if dwelling built in 1960–66 and 0 otherwise | 494.04 | 17.98 |
| $X_7$ | 1 if dwelling built in 1947–59 and 0 otherwise | 334.80 | 11.39 |
| | ($X_5 = X_6 = X_7 = 0$ if dwelling built prior to 1947) | | |
| $X_8$ | 1 if first-story unit in building with elevator and 0 otherwise | —467.04 | —10.63 |
| $X_9$ | 1 if second-story unit in building with elevator and 0 otherwise | —409.20 | —10.86 |
| $X_{10}$ | 1 if third-story unit in building with elevator and 0 otherwise | —347.16 | —9.39 |
| $X_{11}$ | 1 if fourth-story unit in building with elevator and 0 otherwise | —385.68 | —10.54 |
| $X_{12}$ | 1 if fifth-story unit in building with elevator and 0 otherwise | —316.80 | —8.46 |
| $X_{13}$ | 1 if sixth-story unit in building with elevator and 0 otherwise | —371.88 | —9.93 |
| | ($X_8 = \ldots = X_{13} = 0$ if seventh- or higher-story unit in building with elevator) | | |
| $X_{14}$ | 1 if first-story unit in building without elevator and 0 otherwise | —722.88 | —10.30 |
| $X_{15}$ | 1 if second-story unit in building without elevator and 0 otherwise | —826.68 | —11.80 |

TABLE 1 (*Continued*)

| Regressors | Description of Regressors | Coefficients | *t*-Scores |
|---|---|---|---|
| $X_{16}$ | 1 if third-story unit in building without elevator and 0 otherwise | —922.20 | —12.40 |
| $X_{17}$ | 1 if fourth-story unit in building without elevator and 0 otherwise | —923.40 | —7.23 |
| $X_{18}$ | 1 if fifth-story unit in building without elevator and 0 otherwise | —1,304.64 | —8.21 |
| $X_{19}$ | 1 if sixth-story unit in building without elevator and 0 otherwise | —1,627.08 | —6.00 |
| | ($X_{14} = \ldots = X_{19} = 0$ if seventh- or higher-story unit in building without elevator) | | |
| $X_{20}$ | 1 if located in Manhattan and 0 otherwise | 1,174.80 | 24.00 |
| $X_{21}$ | 1 if located in Bronx and 0 otherwise | 104.40 | 2.07 |
| $X_{22}$ | 1 if located in Brooklyn and 0 otherwise | 173.04 | 3.68 |
| $X_{23}$ | 1 if located in Queens and 0 otherwise | 233.76 | 5.05 |
| | ($X_{20} = \ldots = X_{23} = 0$ if located in Richmond) | | |
| $X_{24}$ | 1 if dwelling is in structure with 1–2 units and 0 otherwise | —234.24 | —3.38 |
| $X_{25}$ | 1 if dwelling is in structure with 3–4 units and 0 otherwise | —129.48 | —1.68 |
| $X_{26}$ | 1 if dwelling is in structure with 5–9 units and 0 otherwise | —78.96 | —0.88 |
| $X_{27}$ | 1 if dwelling is in structure with 10–12 units and 0 otherwise | —129.24 | —1.25 |
| $X_{28}$ | 1 if dwelling is in structure with 13–19 units and 0 otherwise | —290.52 | —2.98 |
| $X_{29}$ | 1 if dwelling is in structure with 20–49 units and 0 otherwise | —12.96 | —0.29 |
| $X_{30}$ | 1 if dwelling is in structure with 50–99 units and 0 otherwise | —122.16 | —5.04 |
| | ($X_{24} = \ldots = X_{30} = 0$ if dwelling is in structure with more than 99 units) | | |
| $X_{31}$ | Annual contract rent if apartment is furnished and 0 otherwise | 0.28 | 12.81 |

NOTE.—Dependent variable is annual contract rent. Constant = —61.44; $R^2 = .58$; $s = 653$.

estimate net benefit for each family, the mean net benefit to these families is estimated to have been \$213 during 1968. There is considerable variation around each of these estimates of central tendency. Since there were 1,266,829 families living in controlled housing in 1968, the total benefit to all of these families is estimated to have been about \$270 million. On the basis of our sample, the mean income of families occupying rent controlled housing was \$6,229, so we might reasonably say that rent control resulted in a 3.4 (=\$213/\$6,229) percent increase in their real income. The cost of rent control to the landlords of each controlled apartment is calculated using formula (12), and the mean of these costs is multiplied by the number of controlled apartments to obtain the total cost of rent control to landlords in 1968. This cost was about \$514 million.[17] Dreyfuss and Hendrickson (1968, p. 90) report that the cost of administering the rent-control law in 1968 was about \$7 million. The waste of rent control is simply the excess of the costs borne by landlords and taxpayers over the benefits received by tenants. In 1968, this waste is estimated to have been about \$251 million. These are estimates of some aggregative effects of rent control.[18]

Finally, let us consider some of the distributive effects of rent control. Though there are many rich people living in controlled housing and poor people in uncontrolled housing, table 2 clearly indicates that on average the occupants of rent-controlled apartments are poorer than the occupants of uncontrolled housing. The estimates of the standard deviations of the sample means and proportions reported in parentheses in table 2 suggest that the population parameters are extremely close to their estimates and hence that the differences of means and proportions are significant. Surprisingly, there is no significant difference in the mean age of head of household for the two groups.

We can also investigate the variation in benefits among the families in our sample occupying controlled apartments. As a concise way of presenting this variation, we estimated the following statistical relationships by the method of least squares:

$$\begin{aligned} B = -312.23 &\underset{(-6.48)}{-0.013Y} \underset{(13.48)}{+ 8.63A} \underset{(10.06)}{+ 68.9N} \underset{(1.75)}{+ 44.34R} \underset{(-1.27)}{- 29.31S} \\ R^2 &= .046 \\ s &= 261 \end{aligned} \tag{20}$$

pants of controlled apartments consumed 10.2 percent less housing service in total and 11.7 percent more nonhousing goods (see n. 12 above).

[17] Grampp (1950, p. 436) estimated that the cost of rent control to landlords throughout the United States in 1947 was roughly \$3.87 billion.

[18] The consumers' surplus estimate of mean net tenant benefit is \$405, of total benefit is about \$513.1 million, and of waste is about \$7.9 million. Thus, the use of aggregate data can result in large errors in estimating the benefits and waste from government programs that do not simply reduce the price of a good from some one level to some other level for all subsidized families (see n. 8 above).

TABLE 2

Comparison of Characteristics of Families in Controlled and Uncontrolled Housing

| Characteristics | Controlled | Uncontrolled |
|---|---|---|
| Mean annual income | \$6,223<br>(68) | \$9,000<br>(123) |
| Mean number of members of household | 2.5<br>(0.02) | 2.9<br>(0.02) |
| Mean age of head of household | 49<br>(0.22) | 49<br>(0.24) |
| Percentage of households headed by a nonwhite | 20<br>(0.005) | 17<br>(0.006) |
| Percentage of households headed by a female | 33<br>(0.006) | 20<br>(0.006) |

and

$$\begin{aligned} B/Y = -0.079 - &\underset{(-8.19)}{0.0000056Y} + \underset{(13.8)}{0.003A} + \underset{(3.66)}{0.0086N} + \underset{(2.06)}{0.018R} + \underset{(2.05)}{0.016S} \qquad (21) \\ &R^2 = .051. \\ &s = .261 \end{aligned}$$

The numbers in parentheses are $t$-scores. Under the usual stochastic assumptions, it is clear from (20) that it is quite likely that poorer families receive larger benefits from rent control than richer families, households headed by older persons receive benefits greater than households headed by younger persons, and larger families receive larger benefits than smaller families. Benefits vary less significantly with race and sex of the head of the household. We can be rather certain that blacks receive greater benefits than whites, but we are only moderately confident that households headed by males received larger benefits than households headed by females. A slightly different specification of the stochastic model summarizing the variation in benefit with family characteristics leads to the estimated relationship (21). The same statements can be made about the variation in the percentage increase in real income as were made about the variation in the dollar amount of benefits, except that we can be rather confident that the percentage increase in real income is *less* for households headed by males than for households headed by females.

Therefore, among the set of families who receive a net benefit from rent control, poorer families receive larger benefits. Furthermore, such families are overrepresented in controlled housing. In these senses, rent control achieves some of the objectives desired by supporters of the program. However, the extremely low coefficients of determination in (20) and (21) suggest a great variance in the distribution of benefits among recipient families who are identical in these five characteristics.[19] There is nothing approaching equal treatment of equals among the beneficiaries of rent control. In this sense, rent control is a very poorly focused redistribution device. Unrestricted cash grants or vouchers for particular goods would permit more equal treatment of equally situated families.

## IV. Some Shortcomings

In order to make clear the basic theoretical assumptions, methods of prediction, and empirical results, some potentially important shortcomings of the approach were not mentioned in the preceding sections. The primary purpose of this section is to remedy this defect. In the process, reference will be made to the surprisingly small literature on the economics of rent control.

In this paper it has been assumed that each family is indifferent among all housing units with the same market rent. That is, it has been assumed that all of these dwellings emit the same quantity of housing service per time period. This is an enormously useful simplifying assumption. However, making this assumption precludes considering several of the possibly important sources of inefficiency often attributed to rent control. Specifically, a more detailed theory of the housing market would view housing as a vector of attributes such as space, condition, and location.[20] It has been suggested that the occupants of rent-controlled apartments are not consuming the utility-maximizing combinations of attributes given the market rents of their apartments. In particular, Friedman and Stigler (1946, pp. 15–16), Grampp (1950, pp. 431–33), Grebler (1952, pp. 471–73), and Phelps-Brown and Wiseman (1964, p. 222) argue that most occupants of controlled apartments would prefer less spacious quarters in better condition, while Grampp (1950, pp. 430–31) recognizes that some would prefer lower-quality, larger apartments. Furthermore, it seems likely that many of these families would prefer to live in housing with the same market rent at a different location. As Friedman and Stigler (1946, p. 15), Grampp (1950, pp. 440–41), Paish (1950, p. 7), Turvey (1957, pp. 43–44), and

[19] This characteristic of rent control was hypothesized two decades ago by Johnson (1951, p. 569), who said, "If rent control has been effective in reducing rents, there has been a transfer of real income from landlords to tenants where rents have been controlled. This transfer has been an indiscriminate one, made without regard to the income position of individual tenants." However, there are at least two other factors which contribute to the low coefficient of determination. First, the functional form selected might be far from the mark. Second, even if all of the original data are measured without error, the derived magnitude, net tenant benefit, contains an error component. Therefore, the error term in the regression of estimated net tenant benefit on family characteristics is the sum of the error term in the underlying relationship between net tenant benefit and family characteristics and the error component in the estimation of net tenant benefit. Hence, the variance of the error term in the underlying relationship has probably been overestimated for this reason.

[20] DeSalvo (1970*a*, pp. 6–12) has elaborated the relationship between such a theory and the theory of this paper.

Phelps-Brown and Wiseman (1964, pp. 225–26) point out, rent control creates incentives for people to continue to occupy their controlled apartments even though their places of work change and to reject otherwise preferable jobs far from their present locations. The simple theoretical model of the housing market used in this paper led me to ignore the distorting effect of rent control on the combination of housing characteristics consumed. This is to say that my index of quantity of housing service has the same defect as any index number.[21] As as result, net tenant benefit has probably been overestimated and waste underestimated.

Another source of bias in the same direction stems from the assumption that the controlled rent measures the cost of housing to the occupant of a controlled apartment. In fact, the cost is likely to be higher for several reasons. First, bribes and similar methods are often used by a tenant to gain possession of a controlled apartment (Grampp 1950, p. 437). Although these practices are illegal in New York City (New York Rent and Rehabilitation Administration 1965), there is reason to believe that they occur with some frequency. Second, the time spent searching for an apartment in New York City is probably greater as a result of rent control not only for people who eventually obtain a controlled apartment but also for those who try without success. Third, as Grampp (1950, p. 438) points out, some occupants of controlled housing will find it advantageous to spend money upgrading their apartments. All of these arguments suggest that many occupants of controlled apartments have less to spend on nonhousing goods than my numbers indicate. Hence, the increase in consumption of nonhousing goods has probably been overestimated and for this reason so has net tenant benefit. Some of the extra housing expenditures are transfer payments and have no effect on my estimate of waste. However, at least the greater search costs are real and imply that my estimate of waste is too low. To the extent that these transfer payments accrue to landlords, the cost of rent control to this group has been overstated.

Let us now consider the effect of rent control on the technical efficiency with which housing service is produced in the controlled sector. Paish (1950, pp. 6–7), Grampp (1950, pp. 438–39), Turvey (1957, p. 44), Phelps-Brown and Wiseman (1964, p. 223), and Needleman (1965, p. 164) suggest that, since the landlord's revenue from his rent-controlled apartment is independent of its condition, landlords will not invest in maintenance. Hence, rent-controlled housing will deteriorate faster than it would in the absence of rent control. These authors fail to draw what in my opinion is a potentially important conclusion from this argument, namely, that the cost of producing a unit of housing service in a controlled apartment will be higher than the cost of producing a unit in the uncontrolled market. The argument is as follows. From the viewpoint of tech-

[21] Muth (1960, p. 32) recognized this index number problem; Baumol (1965, pp. 202–5) lucidly explains the problem in the general case.

nical efficiency in the production of housing service, there is an optimal path of deterioration for each dwelling. The operation of a competitive housing market results in the attainment of this optimal path. Rent control results in a faster-than-optimal rate of deterioration. Therefore, rent control results in a higher unit cost of producing housing service in the controlled market. In this paper, it has been assumed that the unit cost of production is the same in the controlled and uncontrolled sectors of the housing market. If the conclusion to the preceding argument is correct, then the cost to the landlord has been underestimated as has the waste due to rent control. It should be noted that the rent-control law in New York City (New York Rent and Rehabilitation Administration 1965) contains many penalties and some rewards to induce the landlord to maintain his controlled apartments. Therefore, we cannot say a priori that controlled housing deteriorates more rapidly than it would in the absence of rent control. However, if rent control has any effect on the rate of deterioration, then the preceding argument need be changed only slightly to reach the same conclusion.

In addition, it seems quite likely that the cost of producing housing service in the uncontrolled market is higher than it would have been had rent control been terminated in New York City shortly after the Second World War. The existence of rent control in New York City probably makes the owners of uncontrolled rental housing sensitive to the possibility of changes in the rent-control law which would inflict capital losses on them. Indeed, in 1969 a mild form of rent control was extended to more than half of the then-uncontrolled rental stock. In other housing markets where rent control was terminated fifteen to twenty years earlier, it is doubtful that landlords attach a very high probability to the imposition of rent control. Therefore, it is reasonable to expect that investors in rental housing in New York City will demand and receive a higher-risk premium. If this argument is correct, then contrary to one of the assumptions underlying the empirical results in this paper, the long-run supply price of housing service in the uncontrolled market in 1968 was greater than it would have been had rent control never gone into effect in New York City. This implies that the occupants of uncontrolled housing were worse off than they would have been in the absence of rent control. It also implies that my predictor of the market rent of controlled housing is biased upward. Hence, the reduction in housing consumption is greater than estimated and net tenant benefit has been overestimated. Likewise, the cost of rent control to landlords has been overestimated. The net effect of this shortcoming on the estimate of waste cannot be determined a priori.

All of these omitted considerations lead me to believe that the aggregate reduction in the consumption of housing service was more than 4.4 percent, that the aggregate increase in consumption of nonhousing service was less than 9.9 percent, that mean net tenant benefit was less than $213, that the total cost to landlords was slightly less than $514 million, and that total waste was more than $251 million. Furthermore, it seems reasonable to conclude that rent control has imposed costs on occupants of uncontrolled apartments.

One of the primary arguments used in behalf of rent control is that landlords are richer than tenants and hence rent control has a desirable redistributive effect. However, if rent control applies to all rental housing in a large area, it is doubtful that the landlords are significantly richer than their tenants. In the only empirical study of this question, Johnson (1951, p. 582) concludes, "I do not want to argue that the evidence presented indicates that landlords are poorer than tenants. But the data certainly do not indicate the contrary—that landlords have significantly higher incomes than tenants. Thus, if one of the objectives of rent control is to aid low-income people—and I can find no other important objective that rent control does achieve—it does not achieve that objective. Undoubtedly, there are relatively poor tenants renting from relatively rich landlords, but the converse must also exist." It is more likely that landlords were richer than tenants in New York City in 1968 than throughout the United States during and immediately after the Second World War, because less than half of all dwellings in New York City were subject to rent control in 1968 and the occupants of these dwellings had a mean income significantly less than the mean income of all families in the city. Unfortunately, the New York City Housing and Vacancy Survey did not collect information that would permit one to relate the costs imposed on landlords by rent control to their characteristics.

Finally, my results show that, among equally situated beneficiaries, benefits are far from equally distributed. If we recognize that for any set of family characteristics there were many families living in uncontrolled housing who received no benefits and perhaps incurred slight costs and a few landlords who incurred large losses from rent control, then we must recognize that the variance of net benefits holding family characteristics constant is probably greatly underestimated by the estimated variance of the error terms in the stochastic models underlying (20) and (21). Thus, expanding our consideration of redistributive effects to the entire population of New York City only reinforces our earlier conclusion. That rent control produces a highly random redistribution of wealth has been suggested by Grampp (1950, p. 447), Paish (1950, p. 4), Johnson (1951, p. 569), Phelps-Brown and Wiseman (1964, pp. 224–25), Needleman (1965, pp. 162–63), and Lindbeck (1967, p. 64).

## References

Aaron, Henry. "Rent Controls and Urban Development: A Case Study of Mexico City." *Soc. and Econ. Studies* 15 (December 1966): 314–28.

Baumol, William J. *Economic Theory and Operations Analysis*. 2d ed. Englewood Cliffs, N.J.: Prentice-Hall, 1965.

de Leeuw, Frank. "The Demand for Housing: A Review of Cross-Section Evidence." *Rev. Econ. and Statis.* 53 (February 1971): 1–10.

de Leeuw, Frank, and Ekanem, Nkanta F. "The Supply of Rental Housing." *A.E.R.* 61 (December 1971): 806–17.

DeSalvo, Joseph S. "A Methodology for Evaluating Housing Programs." P-4364. Santa Monica, Calif.: RAND Corp., 1970. (*a*)

———. "Reforming Rent Control in New York City: Analysis of Housing Expenditures and Market Rentals." *Papers and Proc. Regional Sci. Assoc.* 27 (1970): 195–227. (*b*)

———. "A Methodology for Evaluating Housing Programs." *J. Regional Sci.* 11 (August 1971): 173–85.

Dreyfuss, David, and Hendrickson, Joan. *A Guide to Government Activities in New York City's Housing Markets.* RM-5673-NYC. Santa Monica, Calif.: RAND Corp., 1968.

Fenton, John H. "Boston Returns to Modified Form of Rent Control." *New York Times,* November 22, 1969, p. 21.

Friedman, Milton, and Stigler, George. *Roofs or Ceilings?* Irvington-On-Hudson, N.Y.: Found. Econ. Educ., Inc., 1946.

Goldberger, Arthur S. "The Interpretation and Estimation of Cobb-Douglas Functions." *Econometrica* 35 (July–October 1968): 464–72.

Grampp, William D. "Some Effects of Rent Control." *Southern Econ. J.* 16 (April 1950): 425–47.

Grebler, Leo. "Implications of Rent Control—Experience in the United States." *Internat. Labour Rev.* 65 (April 1952): 462–85.

Hicks, John R. "The Four Consumer Surpluses." *Rev. Economic Studies* 11 (1943): 31–41.

Johnson, D. Gale. "Rent Control and the Distribution of Income." *A.E.R.* 41 (May 1951): 569–82.

Lee, Tong Hun. "The Stock Demand Elasticities of Non-Farm Housing." *Rev. Econ. and Statis.* 46 (February 1964): 82–89.

Lindbeck, Assar. "Rent Control as an Instrument of Housing Policy." In *The Economic Problems of Housing,* edited by Adela Adam Nevitt. London: Macmillan, 1967.

Muth, Richard F. "The Demand for Non-Farm Housing." In *The Demand for Durable Goods,* edited by Arnold C. Harberger. Chicago: Univ. Chicago Press, 1960.

Needleman, Lionel. *The Economics of Housing.* London: Staples, 1965.

New York Rent and Rehabilitation Administration. *The New Little Book on Rent Control.* New York: New York Rent and Rehabilitation Admin., 1965.

Paish, F. W. "The Economics of Rent Restriction." *Lloyd's Bank Rev.* 16 (April 1950): 1–17.

Phelps-Brown, E. H., and Wiseman, J. *A Course in Applied Economics.* 2d ed. London: Pitman & Sons, 1964.

Reid, Margaret G. *Housing and Income.* Chicago: Univ. Chicago Press, 1962.

Shipler, David K. "Cry for Rent Control Grows." *New York Times,* January 12, 1970, p. 35.

Stahl, Ingemar. "Some Aspects of a Mixed Housing Market." In *The Economic Problems of Housing,* edited by Adela Adam Nevitt. London: Macmillan, 1967.

Turvey, Ralph. *The Economics of Real Property.* London: George Allen & Unwin, 1957.

U.S. Bureau of the Census. *Statistical Abstract of the United States: 1970.* 91st ed. Washington: Government Printing Office, 1970.